Beacon

What is a Beacon?

A Beacon is a small physical box that uses Bluetooth Low Energy (BLE) connection. It emits signals at regular intervals which are read directly by a smartphone or tablet when it arrives near the beacons. It is thus possible to send a push notification or trigger an action on the visitor's device.

Beacon sensors are part of geo-location technologies to target users in an established geographic area and deliver content based on where they are. Each beacon allows you to cover a wide field ranging from a few cm up to 450 m, depending on the devices.

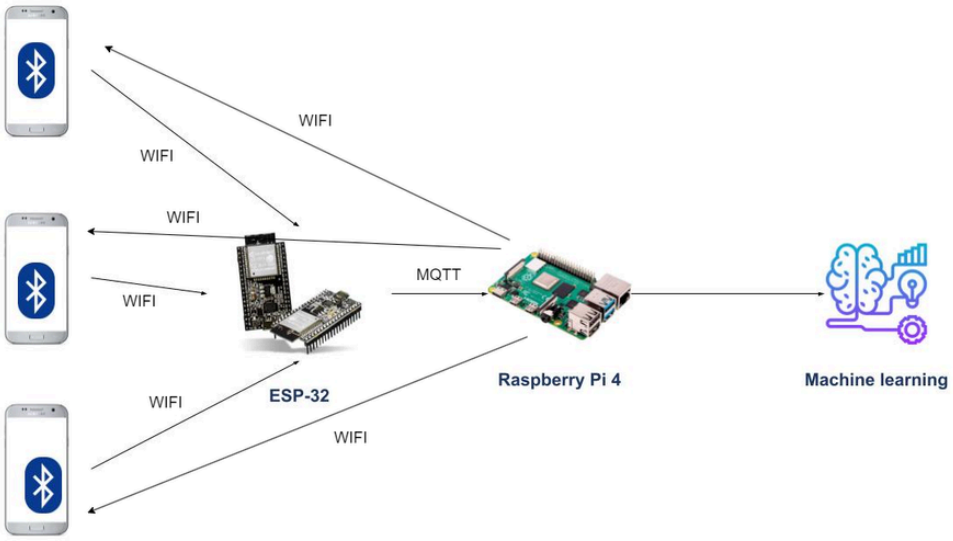
What technology?

BLE stands for Bluetooth Low Energy. It's a power-efficient version of Bluetooth originally introduced in 2010. As BLE don't consume a lot, it allows them to run for years on tiny coin-cell batteries. It also consumes far less energy than the old and clunky Bluetooth. In fact, BLE is a major driver in the IoT, allowing technology to last longer with smaller parts.

Solutions

With Raspberry Pi 4 and ESP32 card

Schema



Material used and properties

Raspberry PI 4 properties:

Double affichage	• Avec le Raspberry Pi 4, vous pouvez exécuter deux moniteurs à la fois et en 4K
Remplacement du PC de bureau	• Pour la première fois, une expérience de bureau complète a été élaborée. Que vous éditiez des documents, naviguiez sur le Web avec de nombreux onglets ouverts, manipuliez des feuilles de calcul, l'expérience serai fluide et reconnaissable. Cependant, vous aurez accès à un format plus petit et plus économe en énergie
Silencieux et faible consommation d'énergie	• Le Raspberry Pi 4 opère silencieusement et dispose d'une consommation d'énergie nettement inférieure à celle des autres ordinateurs
Mise en réseau	• Mini-ordinateur livré avec Ethernet Gigabit, avec réseau sans fil intégré et Bluetooth
USB 3	• Capacité USB améliorée : vous trouverez 2x ports USB 2.0 et 2x ports 3.0 qui peuvent transférer des données jusqu'à dix fois plus rapidement
Choix de RAM	• Différentes variantes en fonction de la quantité de RAM sont disponibles : 1 Go, 2 Go ou 4 Go

Pricing: Raspberry Pi 4 modèle B - Version 1Gb : 38.5€

ESP card properties:

CPU and On-Chip Memory

- ESP32-C6FH4 embedded, 32-bit RISC-V single-core microprocessor, up to 160 MHz
- ROM: 320 KB
- HP SRAM: 512 KB
- LP SRAM: 16 KB
- 4 MB flash in chip package

Wi-Fi

- 1T1R in 2.4 GHz band
- Operating frequency: 2412 ~ 2484 MHz
- IEEE 802.11ax-compliant
  - 20 MHz-only non-AP mode
  - MCS0 ~ MCS9
  - Uplink and downlink OFDMA, especially suitable for simultaneous connections in high-density environments
  - Downlink MU-MIMO (multi-user, multiple input, multiple output) to increase network capacity
  - Beamformee that improves signal quality
  - Channel quality indication (CQI)
  - DCM (dual carrier modulation) to improve link robustness
  - Spatial reuse to maximize parallel transmissions
  - Target wake time (TWT) that optimizes power saving mechanisms
- Fully compatible with IEEE 802.11b/g/n protocol
  - 20 MHz and 40 MHz bandwidth
  - Data rate up to 150 Mbps
  - Wi-Fi Multimedia (WMM)
  - TX/RX A-MPDU, TX/RX A-MSDU
  - Immediate Block ACK
  - Fragmentation and defragmentation
  - Transmit opportunity (TXOP)
  - Automatic Beacon monitoring (hardware TSF)
  - 4 × virtual Wi-Fi interfaces
  - Simultaneous support for Infrastructure BSS in Station mode, SoftAP mode, Station + SoftAP mode, and promiscuous mode *Note that when ESP32-C6 scans in Station mode, the SoftAP channel will change along with the Station channel*
  - 802.11mc FTM

Bluetooth®

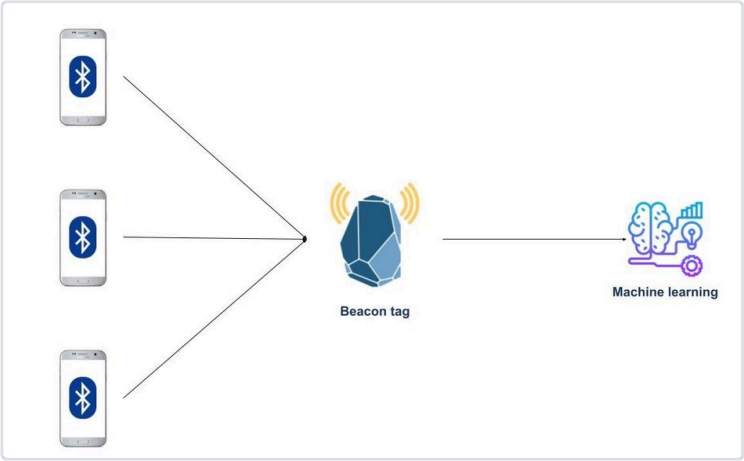
- Bluetooth LE: Bluetooth 5.3 certified
- Bluetooth mesh
- High power mode (20 dBm)
- Speed: 125 Kbps, 500 Kbps, 1 Mbps, 2 Mbps
- Advertising extensions
- Multiple advertisement sets
- Channel selection algorithm #2
- LE power control

For more

Pricing: Carte ESP32-C6-DevKitM-1-N4 : 13,50€

With Beacon

Schema



Material used and properties

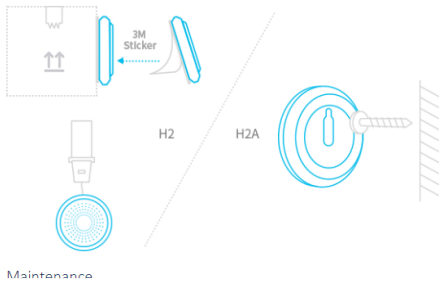
Mokosmart H2 Navigation Beacon properties:

General Features	
Main Chip	Nordic nRF52810
Bluetooth	Bluetooth 4.2 (hardware compatible with bluetooth 5)
Range	Up to 150 meters (in the open area and no obstacles)
Dimension	48.5mm x 48.0mm x 14.7mm(H2) 48.5mm x 48.0mm x 16.3mm(H2A)
Weight	24g (H2) / 25.7g (H2A)
Material	ABS+PC & TPU
Waterproof	IP65
Color	White
Installation	Hang, Neck chain, Sticker
Button	Mechanical button
LED	Single red LED
Sensor	3-axis accelerometer sensor (optional)
Operating temperature	-20°C / + 60°C
Storage temperature	-20°C / + 70°C (without battery) 10°C / + 25°C (with battery)
Humidity	0% ~ 95% (non-condensing)

Functions	
Advertising format	Apple iBeacon Google Eddystone (UID, URL, TLM) MOKO BeaconX Pro sensor data
Multi-advertising	Support up to 6 active slots (Beacon can broadcasts different advertising packets separately)
Configurable list	Advertising format and data/ Beacon name Tx Power/ Advertising interval Connection password/ Trigger options/ Sensor parameters
Sensor sampling	Broadcast or notify the 3-axis accelerometer sensor data in real time
Indoor positioning	Calculate the distance through RSSI and utilize triangulation location to achieve indoor positioning
Configuration APP	iOS and Android are available Scanning the advertising data and connecting to configure all parameters Available APP SDK for customer to develop
Firmware update	Nordic DFU OTA/ J-link

Pricing: Mokosmart H2 Navigation Beacon : 24€

Manual Installation



Youtube link